Chapter 3: Optimal Designs for Gaussian Processes

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1 Introduction

This chapter covers optimal design criteria for Gaussian processes including IMSE, MMSE, and maximum entropy designs.

Citation: V. R. Joseph (2026). *Experimental Design for Data Science and Engineering.* CRC Press.

2 Design Criteria Functions

```
# IMSE (Integrated Mean Squared Error)
imse <- function(D) {
  E <- as.matrix(dist(D))
  R <- exp(-(E / theta)^2)
  L <- chol(R + 10^(-6) * diag(n))
  r <- function(x) exp(-((x - D) / theta)^2)
  s <- function(x) {
    a <- forwardsolve(t(L), r(x))
    return(1 - sum(a^2))
  }
  val <- integrate(Vectorize(s), 0, 1, abs.tol = 10^(-10))$val
  return(val)
}

# MMSE (Maximum Mean Squared Error)
mmse <- function(D) {
  E <- as.matrix(dist(D))
```

```
  R <- exp(-(E / theta)^2)
  L <- chol(R + 10^(-6) * diag(n))
  r <- function(x) exp(-((x - D) / theta)^2)
  s <- function(x) {
    a <- forwardsolve(t(L), r(x))
    return(1 - sum(a^2))
  }
  test <- seq(0, 1, length = 1001)
  val <- apply(cbind(test), 1, s)
  return(max(val))
}

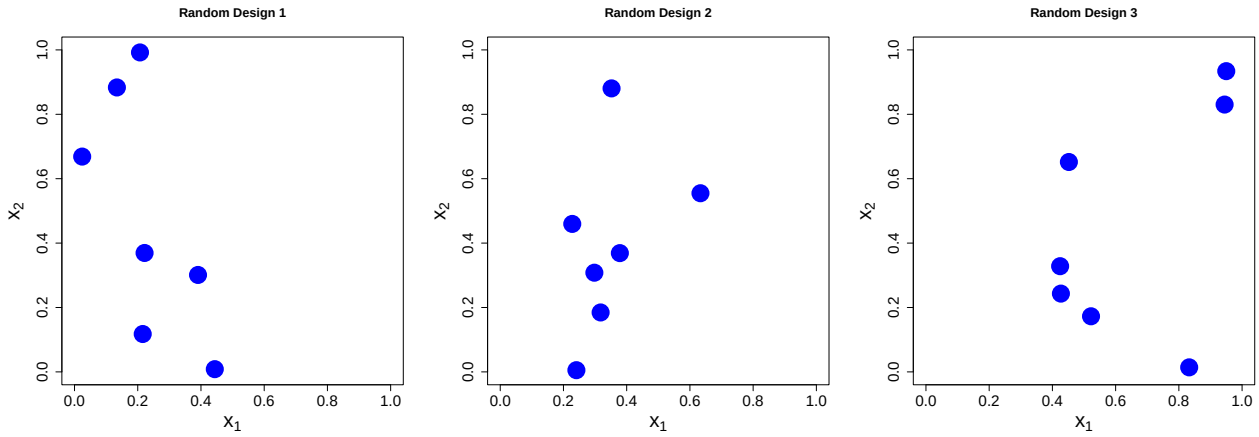
# Entropy
ent <- function(D) {
  E <- as.matrix(dist(D))
  R <- exp(-(E / theta)^2)
  val <- det(R)
  return(val)
}
negent <- function(D) -ent(D)
```

3 Random Design Examples

```
p <- 2
n <- 7

par(mar = c(5, 5, 4, 2))
par(mfrow = c(1, 3))

ind <- c(9, 45, 31)
for (i in 1:3) {
  set.seed(ind[i])
  D <- matrix(runif(n * p), nrow = n, ncol = p)
  plot(D, xlim = c(0, 1), ylim = c(0, 1), xlab = expression(x[1]),
       ylab = expression(x[2]), cex.lab = 2, cex.axis = 1.5,
       pch = 16, col = "blue", cex = 4, main = paste("Random Design", i))
}
```



```r
par(mfrow = c(1, 1))
```

4 RMSE Comparison of Different Designs

```r
n <- 10
theta <- 0.1
N <- 301
test <- seq(0, 1, length = N)

r <- function(x) exp(-((x - D) / theta)^2)
s <- function(x) {
  a <- forwardsolve(t(L), r(x))
  return(sqrt(max(1 - sum(a^2), 0)))
}

par(mfrow = c(2, 2))

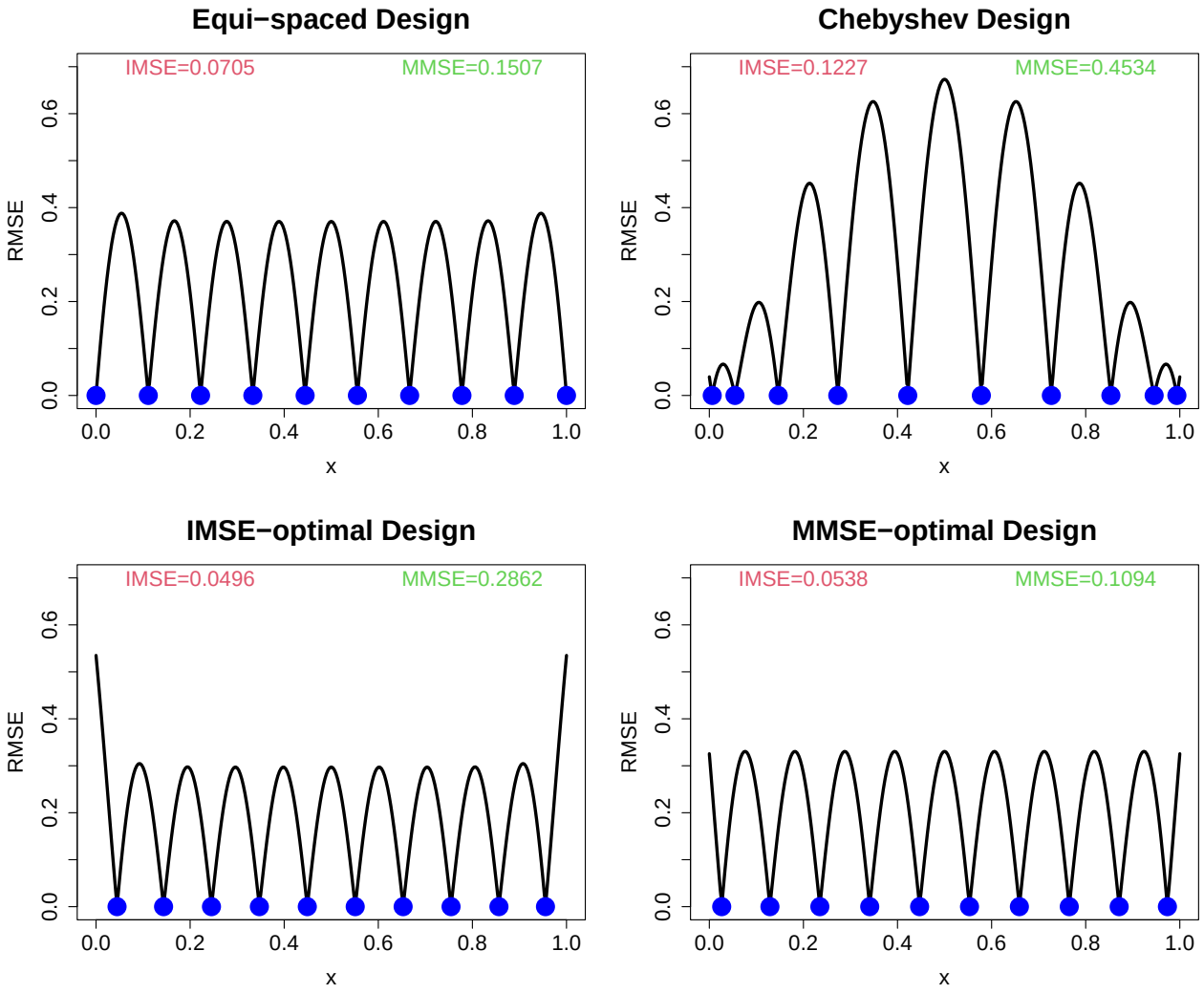
# 1. Equi-spaced design
D <- D1 <- ((1:n) - 1) / (n - 1)
E <- as.matrix(dist(D1))
R <- exp(-(E / theta)^2)
L <- chol(R + 10^(-6) * diag(n))
sa <- apply(cbind(test), 1, s)
plot(test, sa, type = "l", xlab = "x", ylab = "RMSE", ylim = c(0, 0.7),
     main = "Equi-spaced Design", cex.main = 2, cex.lab = 1.5, cex.axis = 1.5, lwd = 3)
points(cbind(D1, 0), pch = 16, col = "blue", cex = 3)
imse_value <- imse(D1)
text(0.2, 0.7, paste0("IMSE=", round(imse_value, 4)), cex = 1.5, col = 2)
mmse_value <- mmse(D1)
text(0.8, 0.7, paste0("MMSE=", round(mmse_value, 4)), cex = 1.5, col = 3)

# 2. Chebyshev design
d <- cos((2 * (1:n) - 1) / (2 * n) * pi)
D <- D2 <- (d + 1) / 2
E <- as.matrix(dist(D2))
R <- exp(-(E / theta)^2)
L <- chol(R + 10^(-6) * diag(n))
sa <- apply(cbind(test), 1, s)
plot(test, sa, type = "l", xlab = "x", ylab = "RMSE", ylim = c(0, 0.7),
     main = "Chebyshev Design", cex.main = 2, cex.lab = 1.5, cex.axis = 1.5, lwd = 3)
points(cbind(D2, 0), pch = 16, col = "blue", cex = 3)
imse_value <- imse(D2)
text(0.2, 0.7, paste0("IMSE=", round(imse_value, 4)), cex = 1.5, col = 2)
mmse_value <- mmse(D2)
text(0.8, 0.7, paste0("MMSE=", round(mmse_value, 4)), cex = 1.5, col = 3)

# 3. IMSE-optimal design
D0 <- ((1:n) - 0.5) / n
a <- optim(D0, imse, lower = rep(0, n), upper = rep(1, n), method = "L-BFGS-B")
D <- D3 <- a$par
E <- as.matrix(dist(D3))
R <- exp(-(E / theta)^2)
```

```r
L <- chol(R + 10^(-6) * diag(n))
sa <- apply(cbind(test), 1, s)
plot(test, sa, type = "l", xlab = "x", ylab = "RMSE", ylim = c(0, 0.7),
     main = "IMSE-optimal Design", cex.main = 2, cex.lab = 1.5, cex.axis = 1.5, lwd = 3)
points(cbind(D3, 0), pch = 16, col = "blue", cex = 3)
imse_value <- imse(D3)
text(0.2, 0.7, paste0("IMSE=", round(imse_value, 4)), cex = 1.5, col = 2)
mmse_value <- mmse(D3)
text(0.8, 0.7, paste0("MMSE=", round(mmse_value, 4)), cex = 1.5, col = 3)

# 4. MMSE-optimal design
a <- optim(D0, mmse, lower = rep(0, n), upper = rep(1, n), method = "L-BFGS-B")
D <- D4 <- a$par
E <- as.matrix(dist(D4))
R <- exp(-(E / theta)^2)
L <- chol(R + 10^(-6) * diag(n))
sa <- apply(cbind(test), 1, s)
plot(test, sa, type = "l", xlab = "x", ylab = "RMSE", ylim = c(0, 0.7),
     main = "MMSE-optimal Design", cex.main = 2, cex.lab = 1.5, cex.axis = 1.5, lwd = 3)
points(cbind(D4, 0), pch = 16, col = "blue", cex = 3)
imse_value <- imse(D4)
text(0.2, 0.7, paste0("IMSE=", round(imse_value, 4)), cex = 1.5, col = 2)
mmse_value <- mmse(D4)
text(0.8, 0.7, paste0("MMSE=", round(mmse_value, 4)), cex = 1.5, col = 3)
```




```
par(mfrow = c(1, 1))
```

5 Relative Efficiency Across Different Theta Values

```
ev <- seq(0.01, 0.4, length = 100)
IR <- MR <- matrix(0, nrow = 100, ncol = 4)

for (i in 1:100) {
  theta <- ev[i]
  IR[i, ] <- c(imse(D1), imse(D2), imse(D3), imse(D4))
  MR[i, ] <- c(mmse(D1), mmse(D2), mmse(D3), mmse(D4))
}

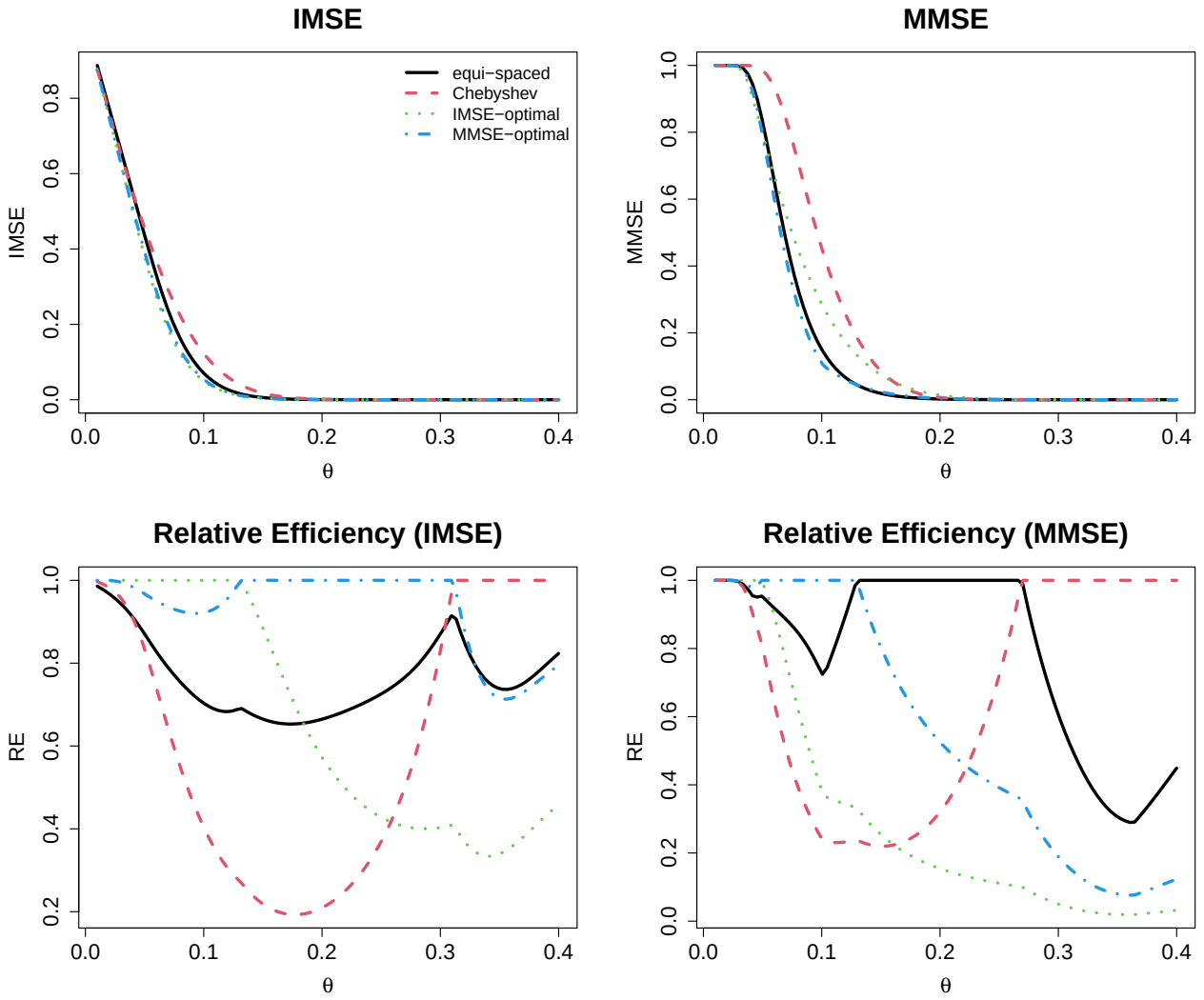
par(mar = c(5, 5, 4, 2))
par(mfrow = c(2, 2))

matplot(ev, IR, type = "l", xlab = expression(theta), ylab = "IMSE", lwd = 3,
        main = "IMSE", cex.main = 2, cex.lab = 1.5, cex.axis = 1.5)
legend("topright", legend = c("equi-spaced", "Chebyshev", "IMSE-optimal", "MMSE-optimal"),
       lty = 1:4, col = 1:4, lwd = 3, bty = "n", cex = 1.2)
```

```r
matplot(ev, MR, type = "l", xlab = expression(theta), ylab = "MMSE", lwd = 3,
        main = "MMSE", cex.main = 2, cex.lab = 1.5, cex.axis = 1.5)

IRR <- sweep(IR, 1, apply(IR, 1, min), "/")
matplot(ev, 1/IRR, type = "l", xlab = expression(theta), ylab = "RE", lwd = 3,
        main = "Relative Efficiency (IMSE)", cex.main = 2, cex.lab = 1.5, cex.axis = 1.5)

MRR <- sweep(MR, 1, apply(MR, 1, min), "/")
matplot(ev, 1/MRR, type = "l", xlab = expression(theta), ylab = "RE", lwd = 3,
        main = "Relative Efficiency (MMSE)", cex.main = 2, cex.lab = 1.5, cex.axis = 1.5)
```




```r
par(mfrow = c(1, 1))
```

6 Maximum Entropy Designs

```r
set.seed(1)
n <- 10
theta <- 0.1
N <- 301
```

```r
test <- seq(0, 1, length = N)

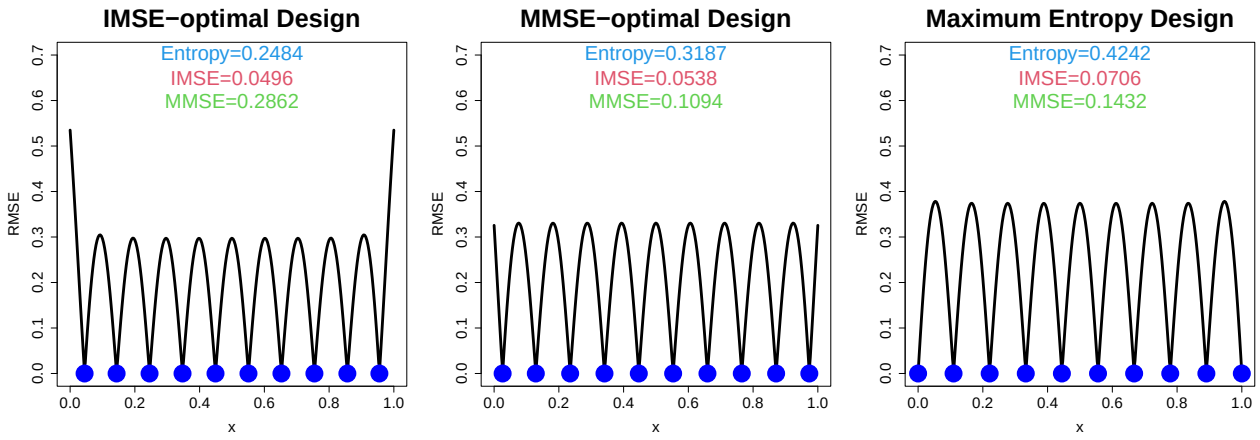
r <- function(x) exp(-((x - D) / theta)^2)
s <- function(x) {
  a <- forwardsolve(t(L), r(x))
  return(sqrt(max(1 - sum(a^2), 0)))
}

par(mfrow = c(1, 3))

# IMSE-optimal design
D0 <- ((1:n) - 0.5) / n
a <- optim(D0, imse, lower = rep(0, n), upper = rep(1, n), method = "L-BFGS-B")
D <- D3 <- a$par
E <- as.matrix(dist(D3))
R <- exp(-(E / theta)^2)
L <- chol(R + 10^(-6) * diag(n))
sa <- apply(cbind(test), 1, s)
plot(test, sa, type = "l", xlab = "x", ylab = "RMSE", ylim = c(0, 0.7),
     main = "IMSE-optimal Design", cex.main = 2.5, cex.lab = 1.5, cex.axis = 1.5, lwd = 3)
points(cbind(D3, 0), pch = 16, col = "blue", cex = 4)
text(0.5, 0.7, paste0("Entropy=", round(ent(D3), 4)), cex = 2, col = 4)
text(0.5, 0.65, paste0("IMSE=", round(imse(D3), 4)), cex = 2, col = 2)
text(0.5, 0.6, paste0("MMSE=", round(mmse(D3), 4)), cex = 2, col = 3)

# MMSE-optimal design
a <- optim(D0, mmse, lower = rep(0, n), upper = rep(1, n), method = "L-BFGS-B")
D <- D4 <- a$par
E <- as.matrix(dist(D4))
R <- exp(-(E / theta)^2)
L <- chol(R + 10^(-6) * diag(n))
sa <- apply(cbind(test), 1, s)
plot(test, sa, type = "l", xlab = "x", ylab = "RMSE", ylim = c(0, 0.7),
     main = "MMSE-optimal Design", cex.main = 2.5, cex.lab = 1.5, cex.axis = 1.5, lwd = 3)
points(cbind(D4, 0), pch = 16, col = "blue", cex = 4)
text(0.5, 0.7, paste0("Entropy=", round(ent(D4), 4)), cex = 2, col = 4)
text(0.5, 0.65, paste0("IMSE=", round(imse(D4), 4)), cex = 2, col = 2)
text(0.5, 0.6, paste0("MMSE=", round(mmse(D4), 4)), cex = 2, col = 3)

# Maximum entropy design
D0 <- ((1:n) - 1) / (n - 1)
a <- optim(D0, negent, lower = rep(0, n), upper = rep(1, n), method = "L-BFGS-B")
D <- D5 <- a$par
E <- as.matrix(dist(D5))
R <- exp(-(E / theta)^2)
L <- chol(R + 10^(-6) * diag(n))
sa <- apply(cbind(test), 1, s)
plot(test, sa, type = "l", xlab = "x", ylab = "RMSE", ylim = c(0, 0.7),
     main = "Maximum Entropy Design", cex.main = 2.5, cex.lab = 1.5, cex.axis = 1.5, lwd = 3)
points(cbind(D5, 0), pch = 16, col = "blue", cex = 4)
text(0.5, 0.7, paste0("Entropy=", round(ent(D5), 4)), cex = 2, col = 4)
text(0.5, 0.65, paste0("IMSE=", round(imse(D5), 4)), cex = 2, col = 2)
text(0.5, 0.6, paste0("MMSE=", round(mmse(D5), 4)), cex = 2, col = 3)
```




```
par(mfrow = c(1, 1))
```

7 Summary

This chapter covered:

1. **IMSE criterion** - Integrated Mean Squared Error for overall prediction accuracy
2. **MMSE criterion** - Maximum Mean Squared Error for worst-case prediction
3. **Maximum entropy designs** - Maximizing information content
4. **Relative efficiency** - Comparing design performance across different lengthscale parameters

Key findings: - Different design criteria lead to different optimal designs - Equi-spaced designs perform reasonably well across many scenarios - Chebyshev designs are optimal for polynomial interpolation - IMSE and MMSE optimal designs differ near boundaries